

Adding a Google App Password for a new application

One 16-character password per application, for michaelbu@mnjlabs.com. Generated 2026-08-04.

What this is (and what it is not)

An **App Password** is a 16-character password that lets a program log in to your Google account without a phone prompt. Programs cannot tap "approve" on your phone, so Google issues this instead.

	The truth
It IS	A second password for your account, usable by any program that has the string.
It is NOT	Restricted to one app. The name you type is only a LABEL for your own bookkeeping - Google does not enforce it.
It is NOT	A key "pair". It is a single string. Nothing to pair it with.
It is NOT	Per-app 2FA. You switch 2-Step Verification on ONCE for the account; it is only the gate that unlocks this page.

Treat it as a full mailbox credential. Anyone holding the string can not only SEND as you - they can READ your whole mailbox over IMAP/POP, bypassing 2FA. That is the point of it. Never commit it, never paste it into a chat, never screenshot it.

So why use one at all? Because it is **independently revocable**: kill that one string and your real password, your phone and your other App Passwords are untouched. If a program used your REAL password and it leaked, the attacker owns the account - and the domain.

Rule: one App Password per application

Do not reuse the Command Center string for a new app. If one leaks you want to revoke it without breaking everything else.

```
michaelbu@mnjlabs.com "MNJ Command Center" -> <16 chars> (existing)
michaelbu@mnjlabs.com "<new app name>" -> <16 chars> (new, different)
```

Name it after the application, exactly as you will remember it in six months. The name is how you know which string to revoke.

Steps

1. Check 2-Step Verification is ON for michaelbu@mnjlabs.com. You did this already - it stays on, you never repeat it per app. (myaccount.google.com/signinoptions/two-step-verification)
2. Go to myaccount.google.com/apppasswords. Sign in again if asked.
3. Type the **new application name** in the "App name" box. Nothing else to choose - there is no app or device dropdown any more.
4. Click **Create**. Google shows 16 characters in four groups, e.g. abcd efgh ijkl mnop.
5. Copy it **now**. Google will never show it again. If you lose it, delete that entry and create another - there is no "reveal".
6. Paste it into the new application config **without the spaces**: abcdefghijklmnop.

7. Restart the application. Environment variables are read at start-up, so a running process never picks up a credential added afterwards.

If the App Passwords page says the option is unavailable, it is one of two things, both admin-side because mnjlabs.com is Google Workspace: (a) Admin console -> Security -> Authentication -> Less secure apps / App passwords is blocked, or (b) the account has Advanced Protection on. You are the domain owner, so you can change it.

Where to put it - the Command Center pattern

Copy this shape for any new app. The secret lives in ONE gitignored file; nothing else holds it.

```
# mail-env.ps1 (gitignored - never commit)
$env:MAIL_USER = "michaelbu@mnjlabs.com"
$env:MAIL_PASS = "abcdefghijklmnop" # the 16 chars, no spaces
$env:MAIL_FROM = "MN&J Labs <michaelbu@mnjlabs.com>"
```

Then add the filename to .gitignore **before** you paste the password in, not after. A secret that reached one commit is in the history for good.

Verify - without sending anything

A real SMTP login that transmits no mail:

```
curl "http://localhost:5178/mail/status?verify=1"
{"configured":true,"verified":true,"user":"michaelbu@mnjlabs.com"}
```

verified: true means the password works. **false** comes with the reason.

Rotating or revoking

1. Go back to `myaccount.google.com/apppasswords`.
2. Delete the old entry by name. It stops working immediately.
3. Create a replacement, update the config file, restart the app.

Rotate whenever the string may have been exposed - a screenshot, a chat, a shared screen, a laptop someone else used. It costs 30 seconds.

If it does not work

Symptom	Cause and fix
MAIL_USER / MAIL_PASS are not set	The app started BEFORE the file existed, or you edited it in a different window. Restart the app.
Username and Password not accepted	Spaces left in the password, or the real account password was used instead of the App Password.
Could not resolve authentication method	The variable is present but EMPTY. An empty value also shadows a real one - check the value, not just that the line exists.
App passwords page unavailable	2-Step Verification is off, or the Workspace admin blocks app passwords, or Advanced Protection is on.

Symptom	Cause and fix
Mail sends but From is wrong	Gmail silently rewrites From to the authenticated mailbox unless the alias is verified in Gmail -> Settings -> Accounts -> Send mail as. Check the RECEIVED header, not the send result.